Describing Distributions & Linear Regression

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Describing shape, center, and spread**

A histogram of household incomes has a long tail stretching toward high values, with most households clustered at lower incomes. Describe the shape of this distribution and state which measure of center (mean or median) is more appropriate to report, with justification.

This distribution is skewed right (the long tail points toward high values). The median is more appropriate, since the mean would be pulled upward by the small number of very high incomes, giving a misleadingly high sense of a "typical" household's income; the median is resistant to this effect.

2 **[Foundational] • Target: Describing shape, center, and spread**

Describe what it means for a distribution to be symmetric, and state the relationship between the mean and median in that case.

A symmetric distribution looks approximately the same on both sides of its center when folded in half. In a symmetric distribution, the mean and median are approximately equal, since there is no long tail pulling the mean away from the median.

3 **[Foundational] • Target: Scatterplots and correlation**

The scatterplot shows hours studied (x) versus test score (y) for 8 students, along with the least-squares regression line $\hat{y} = 52 + 6x$.

a Based on the scatterplot, describe the direction and strength of the association between hours studied and test score. The association is positive (as hours studied increases, test score tends to increase) and appears fairly strong, since the points cluster closely around the upward-sloping line.

b Use the regression equation to predict the test score for a student who studies 4.5 hours.
$\hat{y} = 52 + 6(4.5) = 52 + 27 = 79$.

(a) The association is positive (as hours studied increases, test score tends to increase) and appears fairly strong, since the points cluster closely around the upward-sloping line.; (b)
$\hat{y} = 52 + 6(4.5) = 52 + 27 = 79$.

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4 [Foundational] • Target: Interpreting the correlation coefficient

Explain what a correlation coefficient of $r = 0.85$ indicates about a linear relationship, and explain why r alone cannot confirm that x causes y.

$r = 0.85$ indicates a strong, positive linear relationship between the two variables — as one increases, the other tends to increase as well, and the points cluster fairly tightly around a line. However, correlation does not imply causation: a strong linear association could instead be explained by a lurking (confounding) variable influencing both, or simply by coincidence, so r alone cannot establish that changes in x directly cause changes in y.

5 [Foundational] • Target: Interpreting the correlation coefficient

A data set has correlation coefficient $r = -0.4$. Describe the direction and strength of the linear relationship.

The negative sign indicates a negative association (as x increases, y tends to decrease). The magnitude 0.4 indicates a weak-to-moderate linear relationship, since it is much closer to 0 than to ± 1.

6 [Foundational] • Target: Least-squares regression and residuals

Using the regression equation $\hat{y} = 52 + 6x$ from earlier, find the residual for a student who studied 5 hours and actually scored 79.

Predicted: $\hat{y} = 52 + 6(5) = 82$. Residual $=$ observed $-$ predicted $= 79 - 82 = -3$. The negative residual means this student scored 3 points below what the regression line predicted.

7 [Foundational] • Target: Least-squares regression and residuals

Explain what it means for a residual plot to show "no clear pattern," and why this supports using a linear model for the data.

A residual plot with no clear pattern (residuals scattered randomly above and below zero, with roughly constant spread) suggests that a linear model captures the underlying relationship well — any leftover variation is just random scatter, not a systematic trend the line failed to capture (such as a curve). If a residual plot instead showed a clear curve or fan shape, that would suggest a linear model is not appropriate.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Coefficient of determination

For the study-hours regression, $r = 0.85$. Find r^2 and interpret its meaning in context.

$r^2 = (0.85)^2 = 0.7225$, or about 72.25%. This means approximately 72.25% of the variation in test scores can be explained by the linear relationship with hours studied, leaving about 27.75% of the variation explained by other factors.

9 [Intermediate] • Target: Influential points and outliers in regression

A data point has an unusually large x-value far from the rest of the data, and removing it noticeably changes the slope of the regression line. Explain why this point is called influential, and contrast it with a point that has a large residual but does not change the line much when removed.

A point is influential when its removal substantially changes the regression line (e.g. the slope or intercept) — this typically happens with points that have extreme x-values, since they have high leverage over the line's fit. A point with a large residual but ordinary x-value is simply an outlier in the y-direction; it may look unusual but doesn't have the leverage to substantially shift the line when removed, since points near the middle of the x-range have less influence on slope.

10 [Intermediate] • Target: Extrapolation

Using the regression $\hat{y} = 52 + 6x$ (fit using data from students who studied between 1 and 7 hours), explain why predicting a test score for a student who studied 20 hours would be unreliable.

Predicting for $x = 20$ requires extrapolation — using the regression line far outside the range of x-values (1 to 7 hours) that was actually used to fit it. There is no guarantee the linear relationship continues to hold outside the observed data range (a student's score can't exceed 100, for instance), so such a prediction would likely be unreliable.

11 [Intermediate] • Target: Outliers, Leverage, and Influential Points

A least-squares regression line predicting exam score (y) from homework completion percentage (x) is $\hat{y} = 38.0 + 0.55x$ with correlation $r = 0.82$. (a) Interpret the slope 0.55 in context. (b) Calculate and interpret the coefficient of determination r^2.

(a) Slope Interpretation: For each additional 1 percentage point increase in homework completion, the model predicts an average increase of 0.55 points on the exam. (b) r^2 Calculation & Interpretation: $r^2 = (0.82)^2 = 0.6724 = \mathbf{67.24\%}$. Exactly 67.24% of the variability in exam scores is explained by the linear relationship with homework completion percentage.

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12 **[Intermediate]** • **Target: Describing shape, center, and spread**

Using the regression model $\hat{y} = 38.0 + 0.55x$, a student who completed 80% of homework scored 86 on the exam. (a) Calculate the residual. (b) Explain what a random scatter of points around 0 on a residual plot indicates about model adequacy.

(a) Residual: Predicted score: $\hat{y} = 38.0 + 0.55(80) = 38.0 + 44.0 = 82.0$. Residual: $e = y - \hat{y} = 86 - 82.0 = \mathbf{+4.0}$ points. The student scored 4.0 points higher than predicted by the regression line. (b) Residual Plot Adequacy: A residual plot showing random scatter with no discernible curve, fanning, or systematic pattern confirms that a linear model is appropriate and that error variance is approximately constant.

13 **[Intermediate]** • **Target: Scatterplots and correlation**

A scatterplot is shown below. What is the approximate value of the correlation coefficient, r, for the data presented in the scatterplot?

The scatterplot shows a strong positive linear association between the two variables. As the values of the x-variable increase, the values of the y-variable also tend to increase. The points cluster closely around an upward-sloping line, indicating a strong positive correlation. Therefore, the correlation coefficient, r, should be close to +1. Among the given choices, 0.9 represents the strongest positive linear relationship.

14 **[Intermediate]** • **Target: Interpreting the correlation coefficient**

A researcher is designing a complex experiment to investigate the impact of three different fertilizer treatments (A, B, C) on crop yield, while also controlling for two different irrigation methods (I1, I2) and two soil types (S1, S2). To ensure a robust comparison and account for potential interactions, a stratified randomized block design is employed. Within each unique combination of irrigation method and soil type (forming the blocks), the three fertilizer treatments will be randomly assigned to distinct plots. What is the total number of unique experimental units (plots) that will be established for this experiment?

The problem describes a factorial experimental design with blocking. The blocks are formed by the combinations of irrigation methods and soil types. There are 2 irrigation methods and 2 soil types, so the number of blocks is $2 \times 2 = 4$. Within each block, there are 3 fertilizer treatments that are randomly assigned to plots. Therefore, the number of experimental units (plots) per block is equal to the number of treatments, which is 3. The total number of experimental units is the number of blocks multiplied by the number of units per block. Total units = (Number of irrigation methods) $\times$ (Number of soil types) $\times$ (Number of fertilizer treatments) = $2 \times 2 \times 3 = 12$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging]** • **Target: Least-squares regression and residuals** • **Cross-Lesson: 3.03 The Normal Distribution**

A cereal manufacturer places a collectible card in each box: 20% of boxes have Card A, 30% have Card B, and 50% have Card C. Design a simulation using random digits to estimate the probability that a customer must buy at least 6 boxes to collect all three cards.

1. Assign Digits: Assign single digits proportional to probabilities: Digits 0–1 represent Card A (20%); digits 2–4 represent Card B (30%); digits 5–9 represent Card C (50%). 2. Describe One Trial: Read random digits one at a time. Record which card type is obtained on each digit. Continue reading digits until all three card types (A, B, and C) have appeared at least once. Count the total number of digits (boxes) required. 3. Record Statistic: Record whether the total number of boxes opened is ≥ 6 (Yes/No). 4. Replicate & Estimate: Repeat this trial 100 times. Estimate the probability as the fraction of trials in which 6 or more boxes were required.

16 **[Challenging]** • **Target: Coefficient of determination** • **Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem**

A researcher is designing an experiment to test the effectiveness of a new fertilizer on plant growth. They have a total of 50 experimental plots available. The experiment will involve two treatments: the new fertilizer (Treatment A) and a standard fertilizer (Treatment B). To ensure fair comparison and control for confounding variables, they decide to use a randomized block design. The plots are divided into 5 blocks based on soil quality, with each block containing 10 plots. Within each block, the plots will be randomly assigned to either Treatment A or Treatment B. If the number of plots assigned to Treatment A within the first block is x, and the total number of plots assigned to Treatment A across all 5 blocks is 25, and the proportion of plots assigned to Treatment A in block 1 is $\frac{1}{5}$ of the total plots in block 1, what is the value of x?

The problem describes a randomized block design. There are 5 blocks, and each block has 10 plots, for a total of 50 plots. The experiment uses two treatments: Treatment A (new fertilizer) and Treatment B (standard fertilizer). The total number of plots assigned to Treatment A across all 5 blocks is 25. The question states that the proportion of plots assigned to Treatment A in block 1 is $\frac{1}{5}$ of the total plots in block 1. The number of plots in block 1 is 10. Therefore, the number of plots assigned to Treatment A in block 1, denoted by x, is $x = \frac{1}{5} \times 10 = 2$. Final Answer: 3

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17 **[Challenging]** • **Target: Influential points and outliers in regression** • **Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing**

The commute times for employees at a large tech company have a strongly right-skewed distribution with mean $\mu = 38$ minutes and standard deviation $\sigma = 15$ minutes. A random sample of 49 employees is selected. (a) Describe the shape, mean, and standard error of the sampling distribution of the sample mean $\bar{x}$. (b) Calculate $P(\bar{x} > 42 \text{ minutes})$.

(a) Sampling Distribution Characteristics:Shape: By the Central Limit Theorem, since $n = 49 \geq 30$, the sampling distribution of $\bar{x}$ is approximately normal, despite the right skewness of the population.Mean: $\mu_{\bar{x}} = \mu = \mathbf{38}$ minutes.Standard Error: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{49}} = \frac{15}{7} \approx \mathbf{2.143}$ minutes.(b) Probability Calculation: Standardize: $z = \frac{42 - 38}{2.143} = \frac{4}{2.143} \approx 1.87$. $P(\bar{x} > 42) = P(Z > 1.87) = 1 - 0.9693 = \mathbf{0.0307}$ (3.07%).

18 **[Challenging]** • **Target: Extrapolation** • **Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference**

A data set consists of the following five points:

$P_1 = (2, 3), P_2 = (4, y), P_3 = (6, 7), P_4 = (8, 9), P_5 = (10, 11)$. If the mean of the x-coordinates is 6 and the mean of the y-coordinates is 7, what is the value of y?

The mean of the x-coordinates is given by $\frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = 6$. This matches the information given in the problem. The mean of the y-coordinates is given by $\frac{3 + y + 7 + 9 + 11}{5}$. We are told that the mean of the y-coordinates is 7. Therefore, we can set up the equation: $\frac{3 + y + 7 + 9 + 11}{5} = 7$ Combine the constant terms in the numerator: $\frac{30 + y}{5} = 7$ Multiply both sides by 5: $30 + y = 35$ Subtract 30 from both sides: $y = 5$ Therefore, the value of y is 5.

19 **[Challenging]** • **Target: Interpreting the Least-Squares Regression Line (LSRL)** • **Cross-Lesson: 1.01 Measures of Center & Spread**

A researcher is studying the relationship between study hours and final exam scores for a group of AP Statistics students. The researcher collected data and constructed a scatterplot to visualize the relationship. The scatterplot includes a line of best fit. The equation of the line of best fit is given by $\hat{y} = 2.5x + 15$, where x represents the number of hours studied per week, and $\hat{y}$ represents the predicted final exam score. If a student studied for 12 hours, what is the predicted final exam score for that student?

The equation of the line of best fit is $\hat{y} = 2.5x + 15$. We are given that a student studied for $x = 12$ hours. To find the predicted final exam score, we substitute $x = 12$ into the equation: $\hat{y} = 2.5(12) + 15$. Calculating this gives: $\hat{y} = 30 + 15 = 45$. Therefore, the predicted final exam score is 45. Final Answer: 99

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20 [Challenging] • Target: Residuals and Residual Plots • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

The following diagram shows a scatterplot comparing the number of hours studied per week for a math course and the final exam score for a sample of students. What is the approximate y-intercept of the line of best fit?

The y-intercept of the line of best fit represents the predicted score when the number of hours studied is 0. Looking at the scatterplot, if we were to extend the apparent trend of the data back to the y-axis (where hours studied is 0), the line of best fit would cross the y-axis at approximately 60.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Describing shape, center, and spread

A least-squares regression line is $\hat{y} = 16 - 5x$. Predict $\hat{y}$ when $x = 12$.

a. 76

b. -46

c. -42

d. -44 ✓

$\hat{y} = 16 + (-5)(12) = -44$.

22 [Foundational] • Target: Scatterplots and correlation

A least-squares regression line is $\hat{y} = 82 + 3x$. Predict $\hat{y}$ when $x = 1$.

a. 87

b. 83

c. 79

d. 85 ✓

$\hat{y} = 82 + (3)(1) = 85$.

23 [Foundational] • Target: Interpreting the correlation coefficient

A least-squares regression line is $\hat{y} = 11 + 8x$. Predict $\hat{y}$ when $x = 5$.

a. -29

b. 49

c. 53

d. 51 ✓

$\hat{y} = 11 + (8)(5) = 51$.

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24 [Foundational] • Target: Least-squares regression and residuals

A least-squares regression line is $\hat{y} = 55 + 3x$. Predict $\hat{y}$ when $x = 13$.

a. 16

b. 96

c. 94 ✓

d. 92

$\hat{y} = 55 + (3)(13) = 94$.

25 [Foundational] • Target: Coefficient of determination

A least-squares regression line is $\hat{y} = 32 - 4x$. Predict $\hat{y}$ when $x = 15$.

a. -26

b. -30

c. -28 ✓

d. 92

$\hat{y} = 32 + (-4)(15) = -28$.

26 [Foundational] • Target: Influential points and outliers in regression

A least-squares regression line is $\hat{y} = 39 - 5x$. Predict $\hat{y}$ when $x = 18$.

a. -51 ✓

b. -49

c. 129

d. -53

$\hat{y} = 39 + (-5)(18) = -51$.

27 [Foundational] • Target: Extrapolation

A least-squares regression line is $\hat{y} = 79 + 4x$. Predict $\hat{y}$ when $x = 1$.

a. 75

b. 83 ✓

c. 85

d. 81

$\hat{y} = 79 + (4)(1) = 83$.

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28 [Foundational] • Target: Interpreting the Least-Squares Regression Line (LSRL)

A least-squares regression line is $\hat{y} = 52 + 2x$. For a point with $x = 2$ and observed value $y = 49$, find the residual (observed − predicted).

a. -9

b. -5

c. 7

d. -7 ✓

Predicted: $\hat{y} = 52 + (2)(2) = 56$. Residual $= y - \hat{y} = 49 - (56) = -7$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Residuals and Residual Plots

A least-squares regression line is $\hat{y} = 73 - 9x$. For a point with $x = 6$ and observed value $y = 29$, find the residual (observed − predicted).

a. 8

b. 12

c. 10 ✓

d. -10

Predicted: $\hat{y} = 73 + (-9)(6) = 19$. Residual $= y - \hat{y} = 29 - (19) = 10$.

30 [Intermediate] • Target: Correlation Coefficient r and Coefficient of Determination r^2

A least-squares regression line is $\hat{y} = 36 - 3x$. For a point with $x = 13$ and observed value $y = 1$, find the residual (observed − predicted).

a. 2

b. 4 ✓

c. -4

d. 6

Predicted: $\hat{y} = 36 + (-3)(13) = -3$. Residual $= y - \hat{y} = 1 - (-3) = 4$.

31 [Intermediate] • Target: Outliers, Leverage, and Influential Points

A least-squares regression line is $\hat{y} = 26 + 7x$. For a point with $x = 2$ and observed value $y = 41$, find the residual (observed − predicted).

a. 3

b. 1 ✓

c. 4

d. -1

Predicted: $\hat{y} = 26 + (7)(2) = 40$. Residual $= y - \hat{y} = 41 - (40) = 1$.

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32 [Intermediate] • Target: Describing shape, center, and spread

A least-squares regression line is $\hat{y} = 21 - 2x$. For a point with $x = 10$ and observed value $y = -4$, find the residual (observed − predicted).

a. -5 ✓

b. -3

c. 5

d. -7

Predicted: $\hat{y} = 21 + (-2)(10) = 1$. Residual $= y - \hat{y} = -4 - (1) = -5$.

33 [Intermediate] • Target: Scatterplots and correlation

A least-squares regression line is $\hat{y} = 64 + 7x$. For a point with $x = 5$ and observed value $y = 101$, find the residual (observed − predicted).

a. 4

b. 0

c. 2 ✓

d. -2

Predicted: $\hat{y} = 64 + (7)(5) = 99$. Residual $= y - \hat{y} = 101 - (99) = 2$.

34 [Intermediate] • Target: Interpreting the correlation coefficient

A least-squares regression line is $\hat{y} = 45 - 9x$. For a point with $x = 15$ and observed value $y = -91$, find the residual (observed − predicted).

a. -1 ✓

b. 1

c. -4

d. -3

Predicted: $\hat{y} = 45 + (-9)(15) = -90$. Residual $= y - \hat{y} = -91 - (-90) = -1$.

35 [Intermediate] • Target: Least-squares regression and residuals

A regression has correlation coefficient $r = -0.65$. Find r^2, expressed as a percentage, to two decimal places.

a. 42.25% ✓

b. 44.25%

c. 40.25%

d. 65.00%

$r^2 = (-0.65)^2 = 0.4225$, or 42.25%.

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36 [Intermediate] • Target: Coefficient of determination

A regression has correlation coefficient $r = 0.50$. Find r^2, expressed as a percentage, to two decimal places.

a. 50.00%

b. 25.00% ✓

c. 27.00%

d. 23.00%

$r^2 = (0.50)^2 = 0.2500$, or 25.00%.

37 [Intermediate] • Target: Influential points and outliers in regression

A distribution has mean 94 and median 82. Based on this comparison, classify the shape of the distribution.

a. Approximately symmetric (mean and median are close)

b. **Skewed right (mean noticeably greater than median)** ✓

c. Skewed left (mean noticeably less than median)

d. Cannot be determined from the mean and median alone

Mean $= 94$, median $= 82$. Skewed right (mean noticeably greater than median).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Extrapolation • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A distribution has mean 83 and median 84. Based on this comparison, classify the shape of the distribution.

a. **Approximately symmetric (mean and median are close)** ✓

b. Skewed right (mean noticeably greater than median)

c. Skewed left (mean noticeably less than median)

d. Cannot be determined from the mean and median alone

Mean $= 83$, median $= 84$. Approximately symmetric (mean and median are close).

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39 [Challenging] • Target: Interpreting the Least-Squares Regression Line (LSRL) • Cross-Lesson: 1.01 Measures of Center & Spread

A regression has coefficient of determination $r^2 = 0.64$. Find the possible value(s) of the correlation coefficient r.

a. ± 0.64

b. 0.80

c. ± 0.90

d. ± 0.80 ✓

$r = \pm\sqrt{0.64} = \pm 0.80$ (since r^2 does not determine the sign of r, both the positive and negative square root are possible).

40 [Challenging] • Target: Residuals and Residual Plots • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A regression has coefficient of determination $r^2 = 0.16$. Find the possible value(s) of the correlation coefficient r.

a. ± 0.16

b. ± 0.50

c. 0.40

d. ± 0.40 ✓

$r = \pm\sqrt{0.16} = \pm 0.40$ (since r^2 does not determine the sign of r, both the positive and negative square root are possible).

41 [Challenging] • Target: Interpreting the Least-Squares Regression Line (LSRL) • Cross-Lesson: 2.01 Sampling Methods & Bias

A regression model predicting used car price in dollars from age in years is $\hat{y} = 24000 - 1800x$. Interpret the slope -1800 in context.

a. **For each additional year of car age, the predicted price decreases by 1,800 dollars** ✓

b. A car loses exactly 1,800 dollars every year

c. When the car is 0 years old, its predicted price is $-1,800$ dollars

d. The correlation between age and price is -1800

The slope of an LSRL indicates the predicted (or estimated average) change in the response variable (y) for each 1-unit increase in the explanatory variable (x).

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42 [Challenging] • Target: Interpreting the Least-Squares Regression Line (LSRL) • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Why is the line called the 'least-squares' regression line?

a. It minimizes the sum of the squared vertical residuals $\sum(y - \hat{y})^2$ ✓

b. It minimizes the sum of squared distances to the origin

c. It maximizes the square of the correlation coefficient

d. It guarantees all residuals are perfect squares

The LSRL is mathematically derived to minimize the sum of squared vertical differences between observed and predicted values.

43 [Challenging] • Target: Residuals and Residual Plots • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

Which pattern in a residual plot indicates that a linear model is appropriate?

a. A random scatter of points with no discernible pattern around the residual $= 0$ line ✓

b. A clear curved U-shape pattern

c. A fan shape where residuals expand as x increases

d. Points clustered strictly along an inclined line

A random scatter of residuals around 0 with constant vertical spread indicates that a linear relationship is suitable for the data.

44 [Challenging] • Target: Correlation Coefficient r and Coefficient of Determination r^2 • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

What is the correct statistical interpretation of $r^2 = 0.75$?

a. 75% of the variation in the response variable y is explained by the linear relationship with x ✓

b. 75% of the data points lie directly on the regression line

c. The slope of the regression line is 0.75

d. There is a 75% probability that x causes y

r^2 measures the proportion of variability in the response variable that is accounted for by the linear model with the explanatory variable.

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45 [Challenging] • Target: Outliers, Leverage, and Influential Points • Cross-Lesson: 3.03 The Normal Distribution

A point that has an extreme x-value relative to the rest of the data is said to have:

a. **High leverage ✓**

b. A large residual

c. High correlation

d. Negative slope

Observations with extreme explanatory (x) values possess high leverage, giving them significant potential to alter the slope of the regression line.

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